ClipGS: Clippable Gaussian Splatting

For Interactive Cinematic Visualization of Volumetric Medial Data

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Medizinische Bild- und Signalverarbeitung

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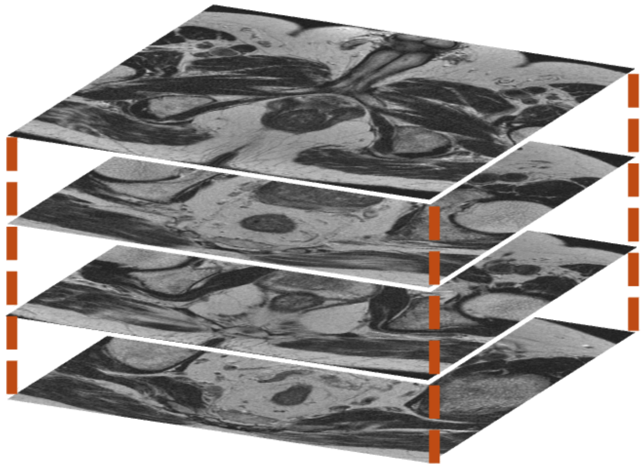
Scenario

Input

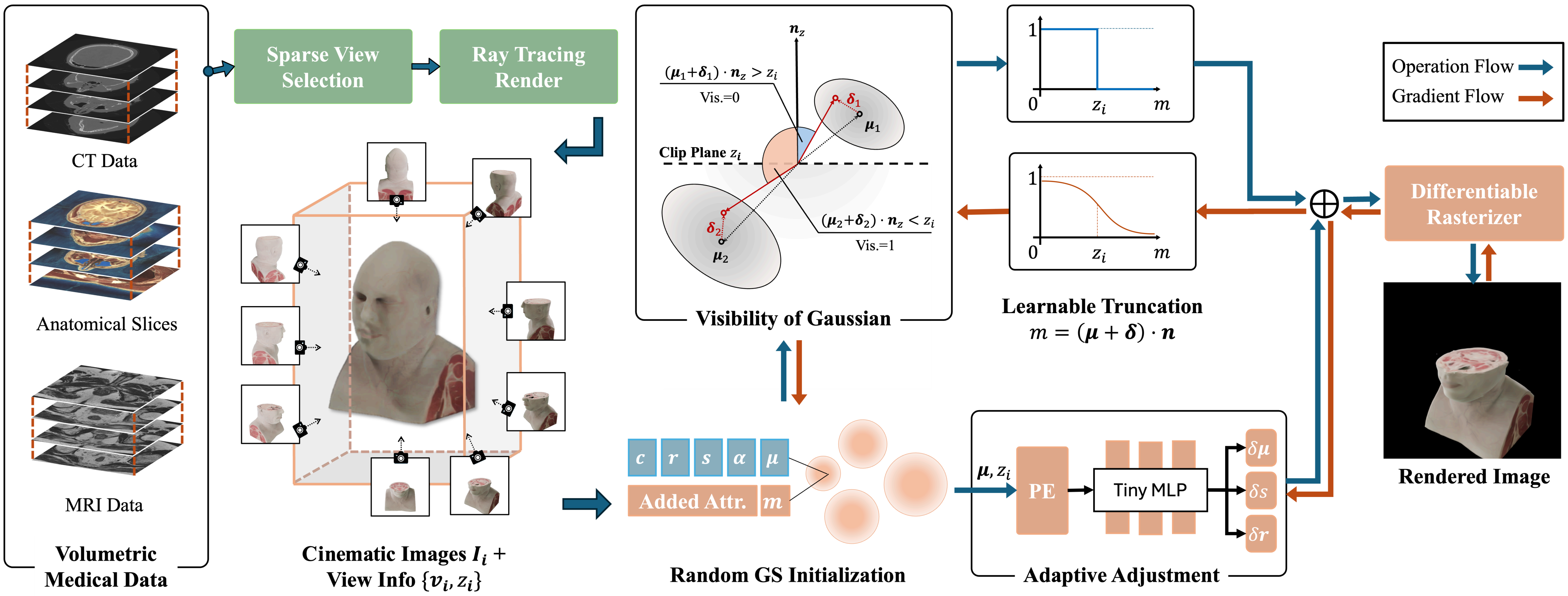


Volumetric Medical Data

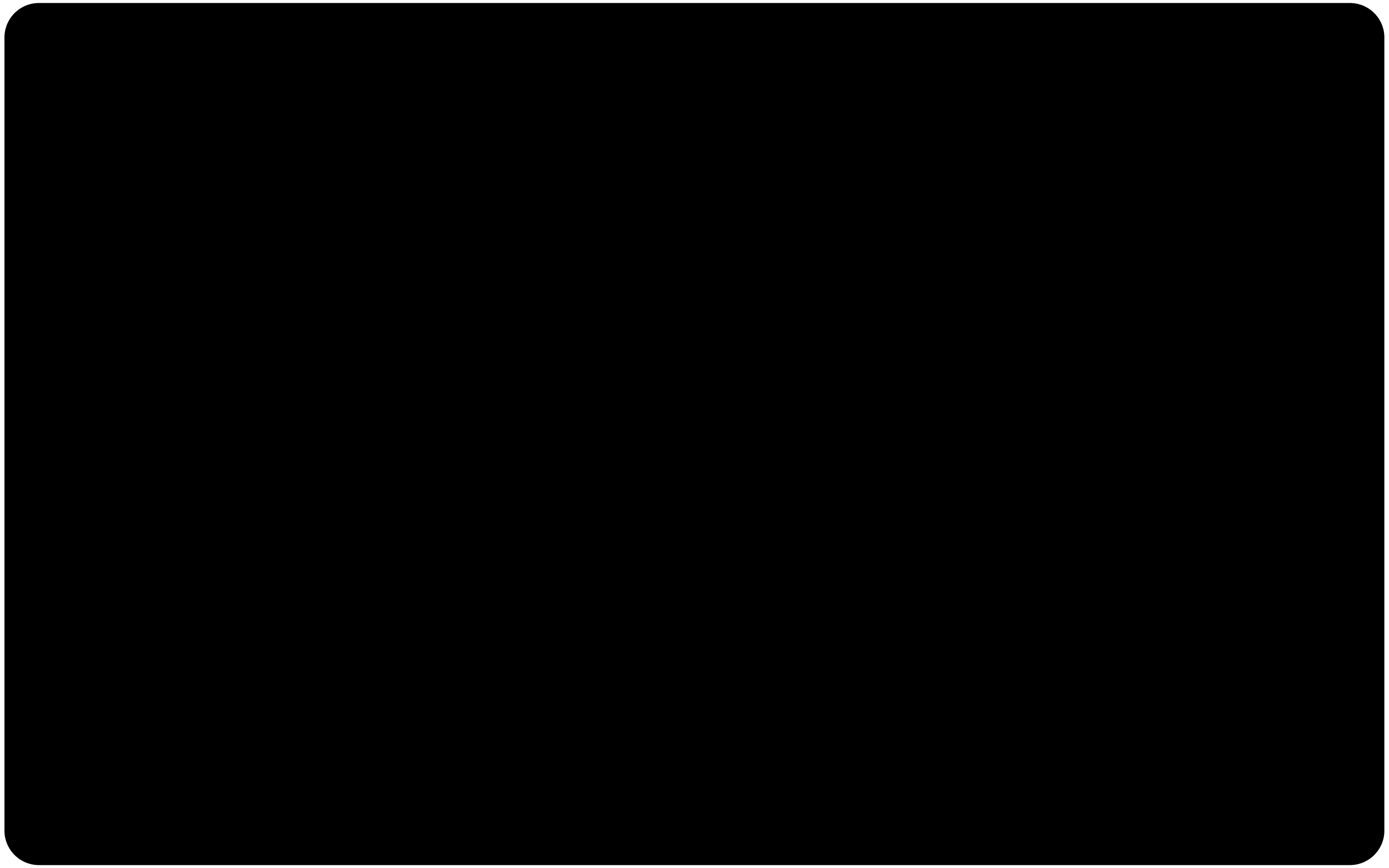
Pathtraced Render

- View $\mathbf{v}_i$
- Clip Plane Depth z_i
- Not Interactive

Method Preview



3D Gaussian Splatting (3DGS)



gsplat.js renderer

Gaussian Parameters

$$\alpha \cdot \mathcal{G}(\mathbf{x}; \boldsymbol{\mu}, \mathbf{s}, \mathbf{r}) \cdot \mathbf{c}$$

Position $\boldsymbol{\mu}$

x

y

z

Scale $\mathbf{s}$

x

y

z

Rotation $\mathbf{r}$

x

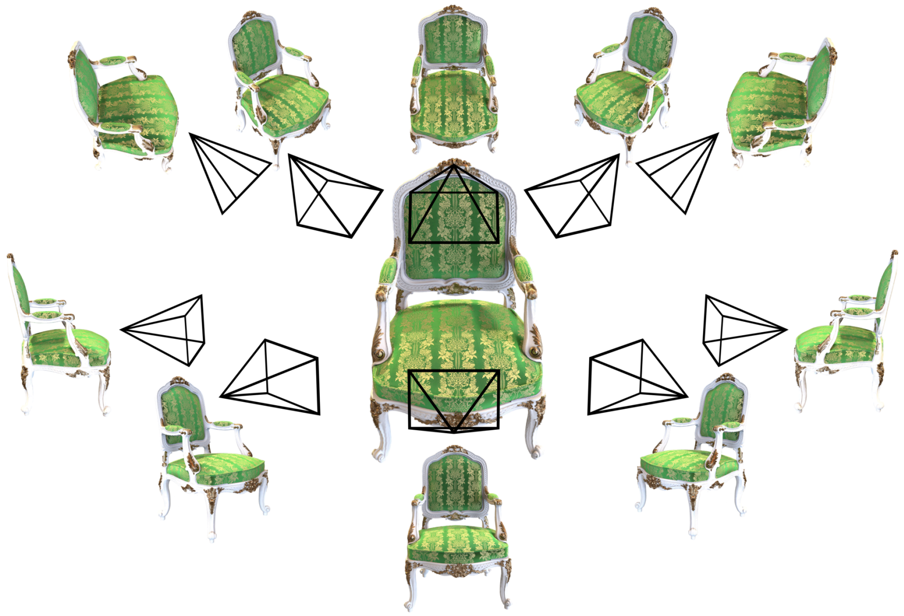
y

z

Color $\mathbf{c}$

Opacity α

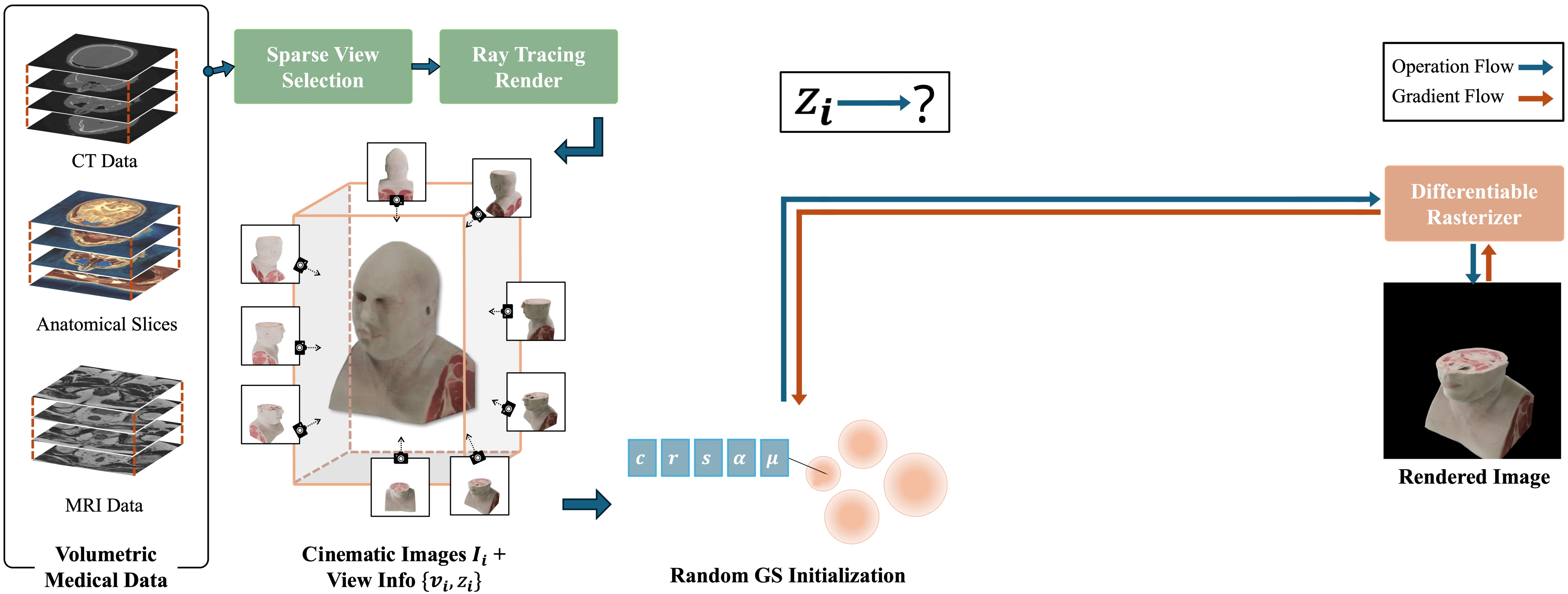
3DGS Training



Renders created with PlenOctrees

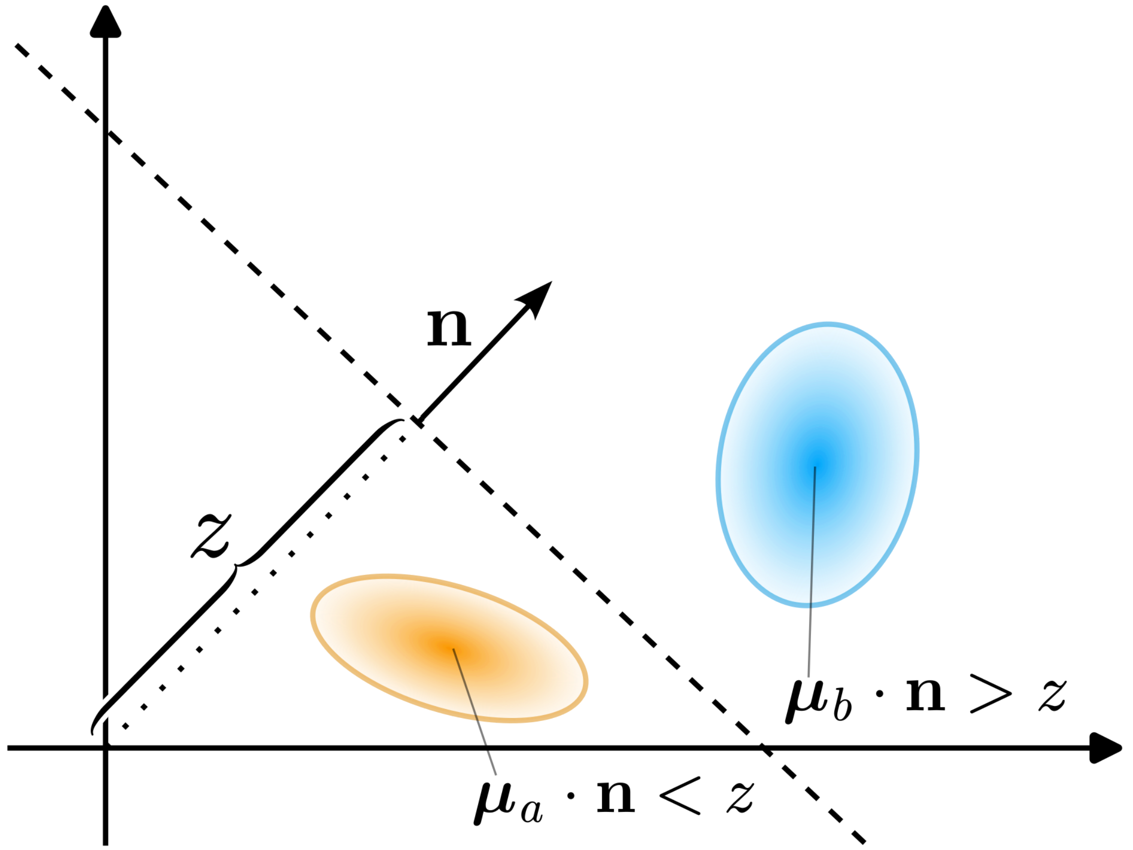
Lecture: Computer Graphics — Mario Botsch

Base Method

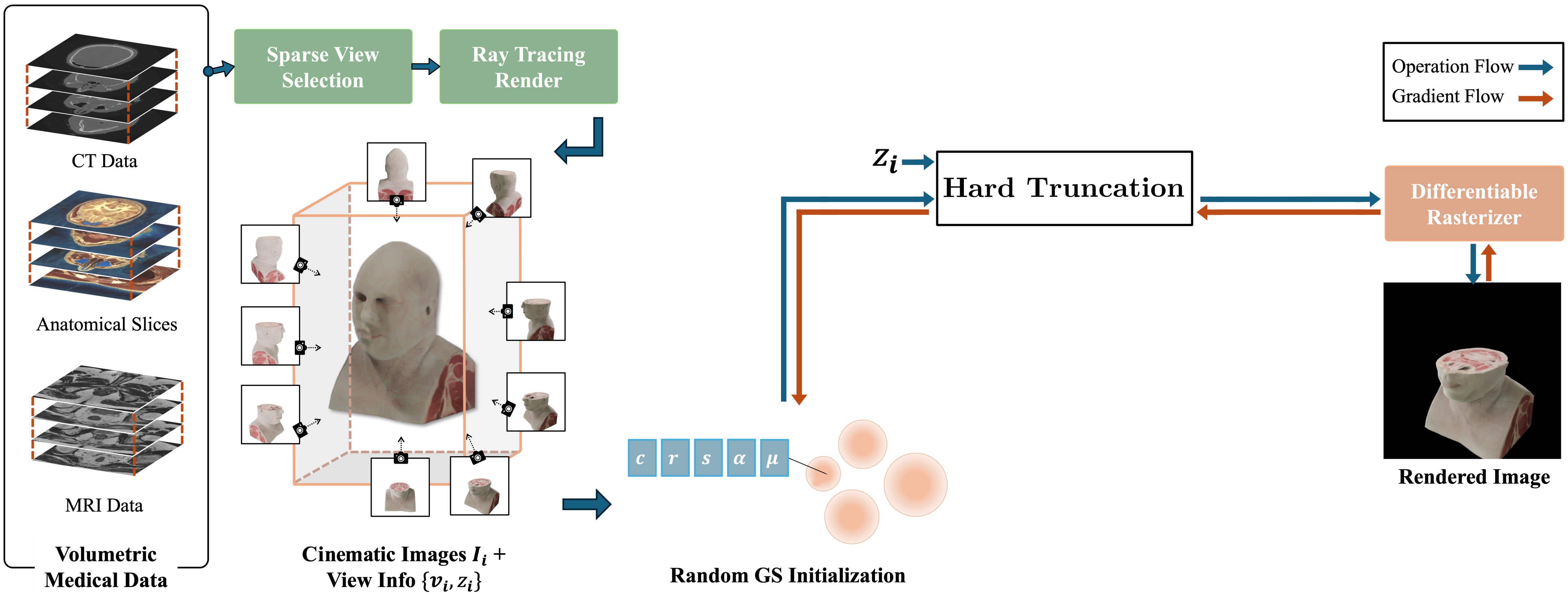


CT Data
Anatomical Slices
MRI Data
Volumetric Medical Data
Sparse View Selection
Ray Tracing Render
Cinematic Images I_i + View Info $\{v_i, z_i\}$
c r s α μ
Random GS Initialization
$z_i \longrightarrow ?$
Operation Flow
Gradient Flow
Differentiable Rasterizer
Rendered Image

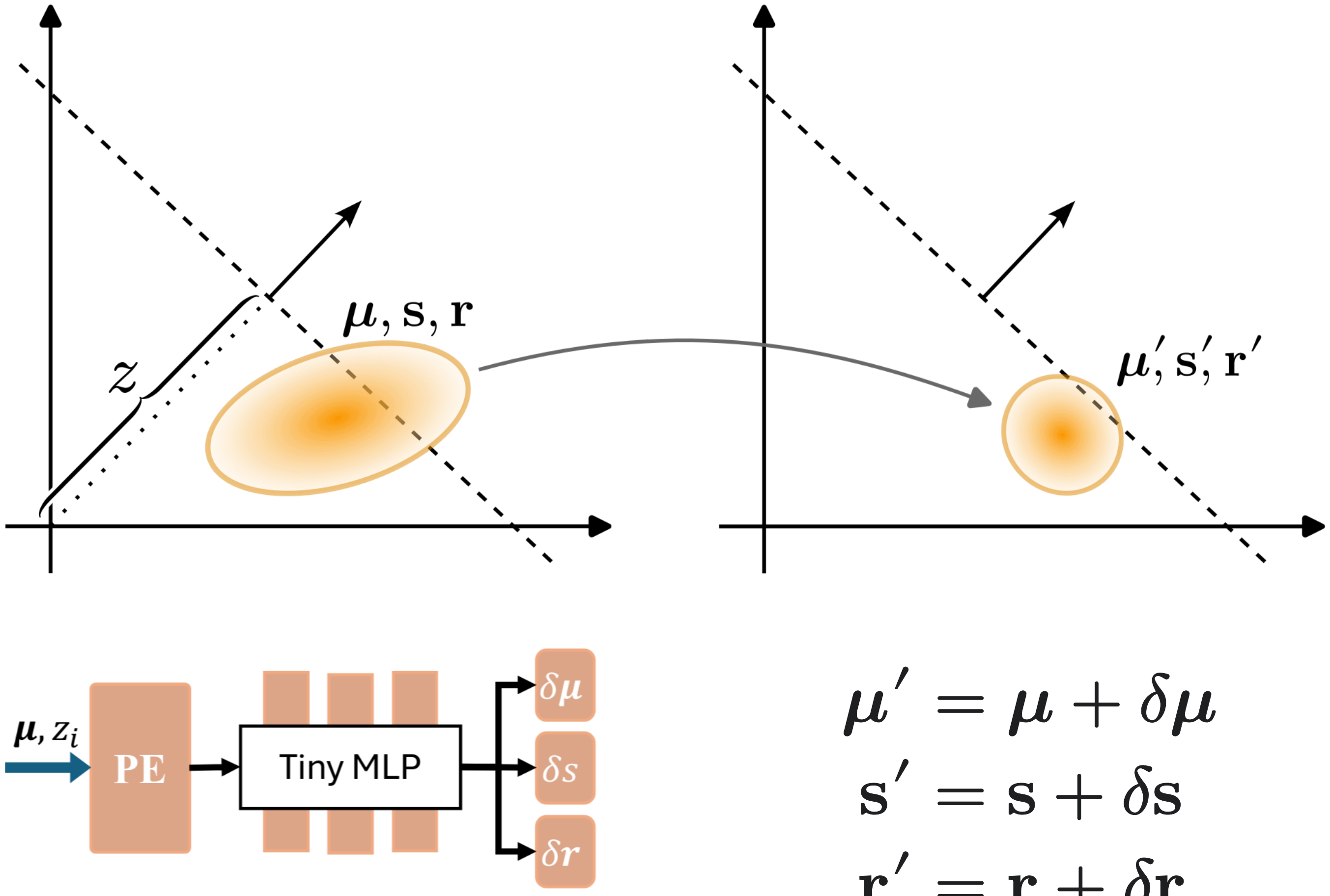
Hard Truncation (HT)



Hard Truncation Method



Adaptive Adjustment Model (AAM)

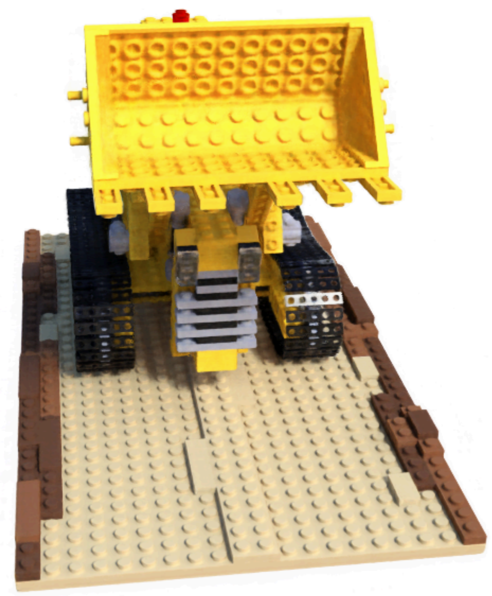


$$\boldsymbol{\mu}' = \boldsymbol{\mu} + \delta\boldsymbol{\mu}$$
$$\mathbf{s}' = \mathbf{s} + \delta\mathbf{s}$$
$$\mathbf{r}' = \mathbf{r} + \delta\mathbf{r}$$

Positional Encoding (PE)

$$\gamma(x) = [\sin(2^0 \pi x), \cos(2^0 \pi x), \dots, \sin(2^L \pi x), \cos(2^L \pi x)]^\top$$

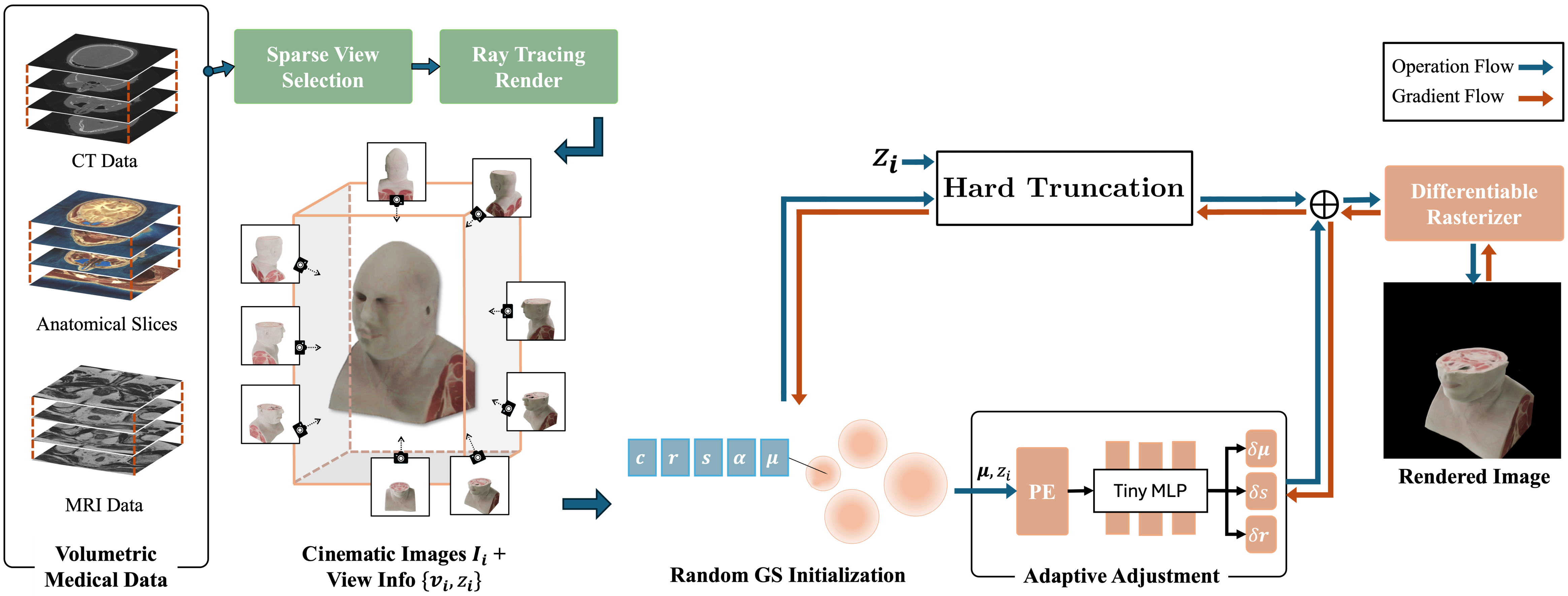
Example – NeRF



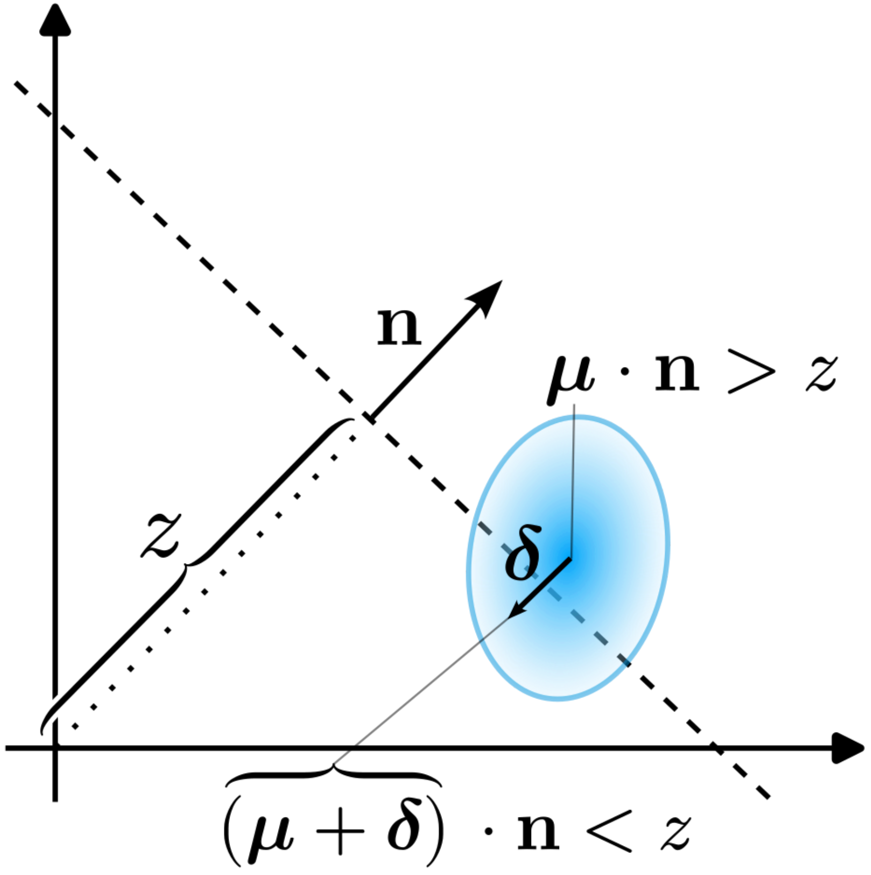
With PE

Without PE

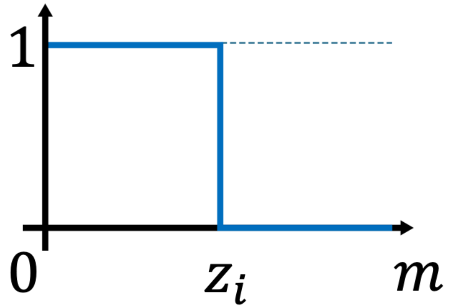
HT + AAM Method



Learned Truncation

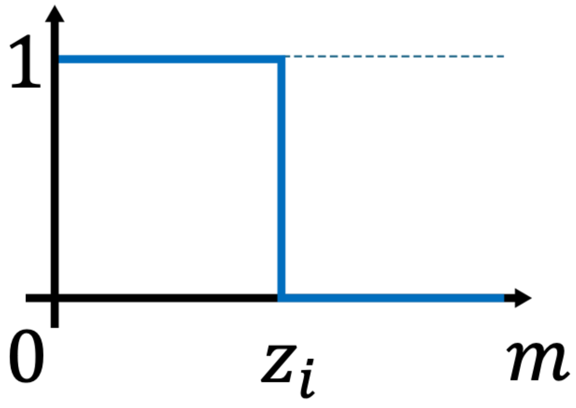


- $\cancel{(\boldsymbol{\mu} + \boldsymbol{\delta}) \cdot \mathbf{n}} < z$
- $m < z$
  - One degree of freedom necessary
  - Decouple clipping and position
- Not differentiable

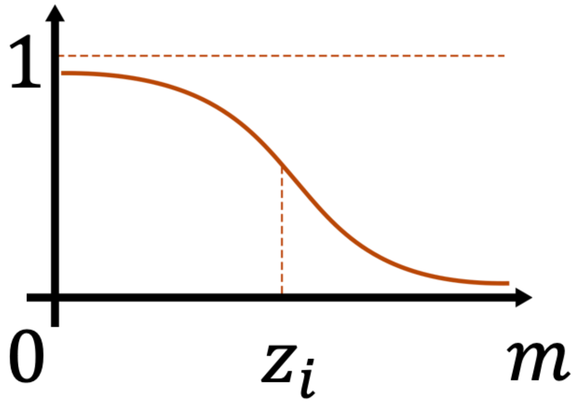


Straight Through Estimator

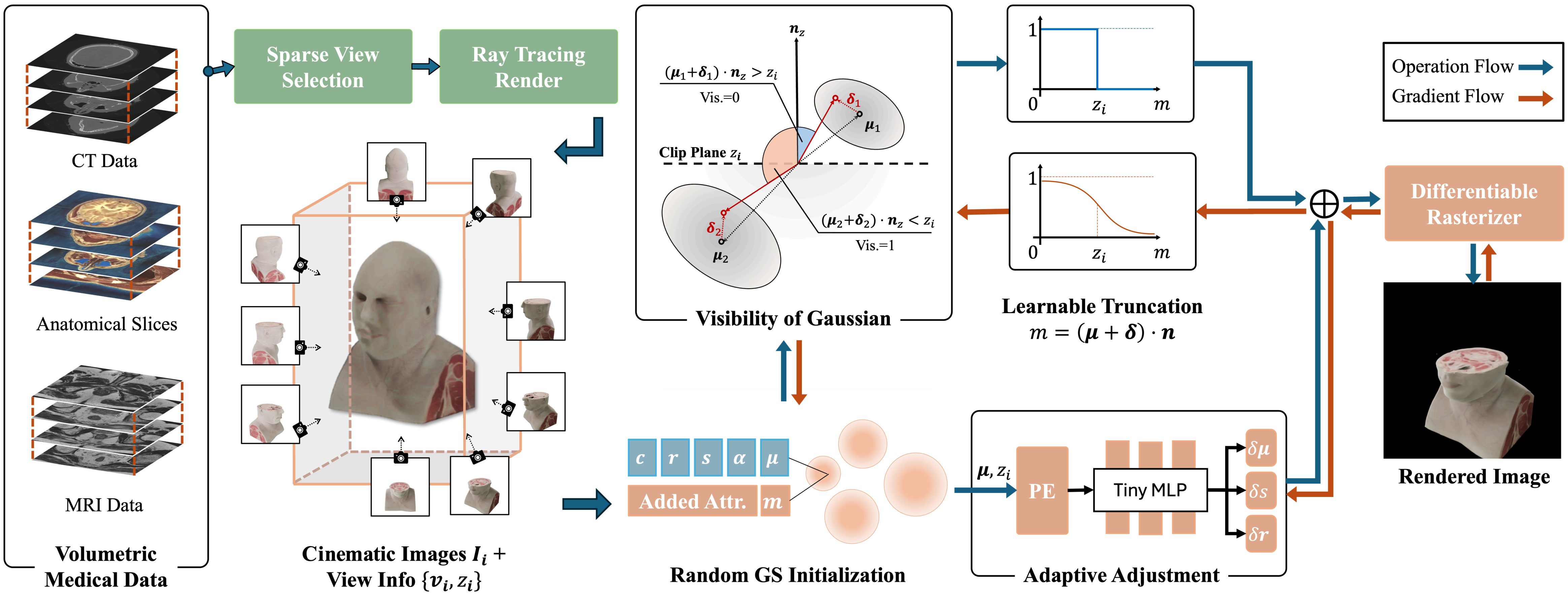
Forward Pass



Backward Pass



Full Method



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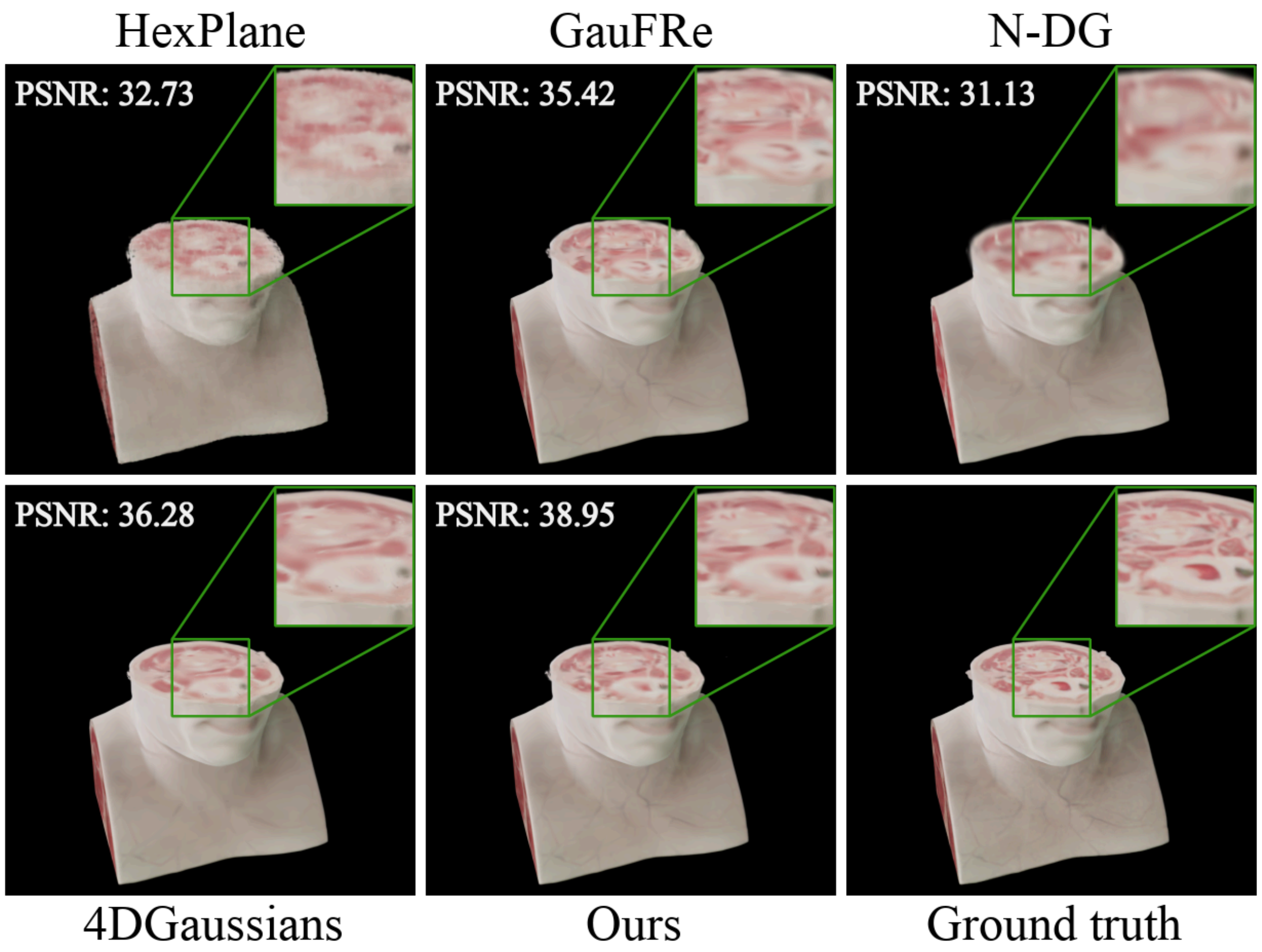
Sparse View Selection
Ray Tracing Render
Cinematic Images I_i + View Info $\{v_i, z_i\}$
n_z
$(\mu_1+\delta_1) \cdot n_z > z_i$
Vis.=0
Clip Plane z_i
δ_1
μ_1
δ_2
μ_2
$(\mu_2+\delta_2) \cdot n_z < z_i$
Vis.=1
Visibility of Gaussian
1
0
z_i
m
Learnable Truncation
$m = (\mu + \delta) \cdot n$
Operation Flow
Gradient Flow
Differentiable Rasterizer
Rendered Image
c r s α μ
Added Attr. m
Random GS Initialization
μ, z_i
PE
Tiny MLP
$\delta\mu$
δs
δr
Adaptive Adjustment

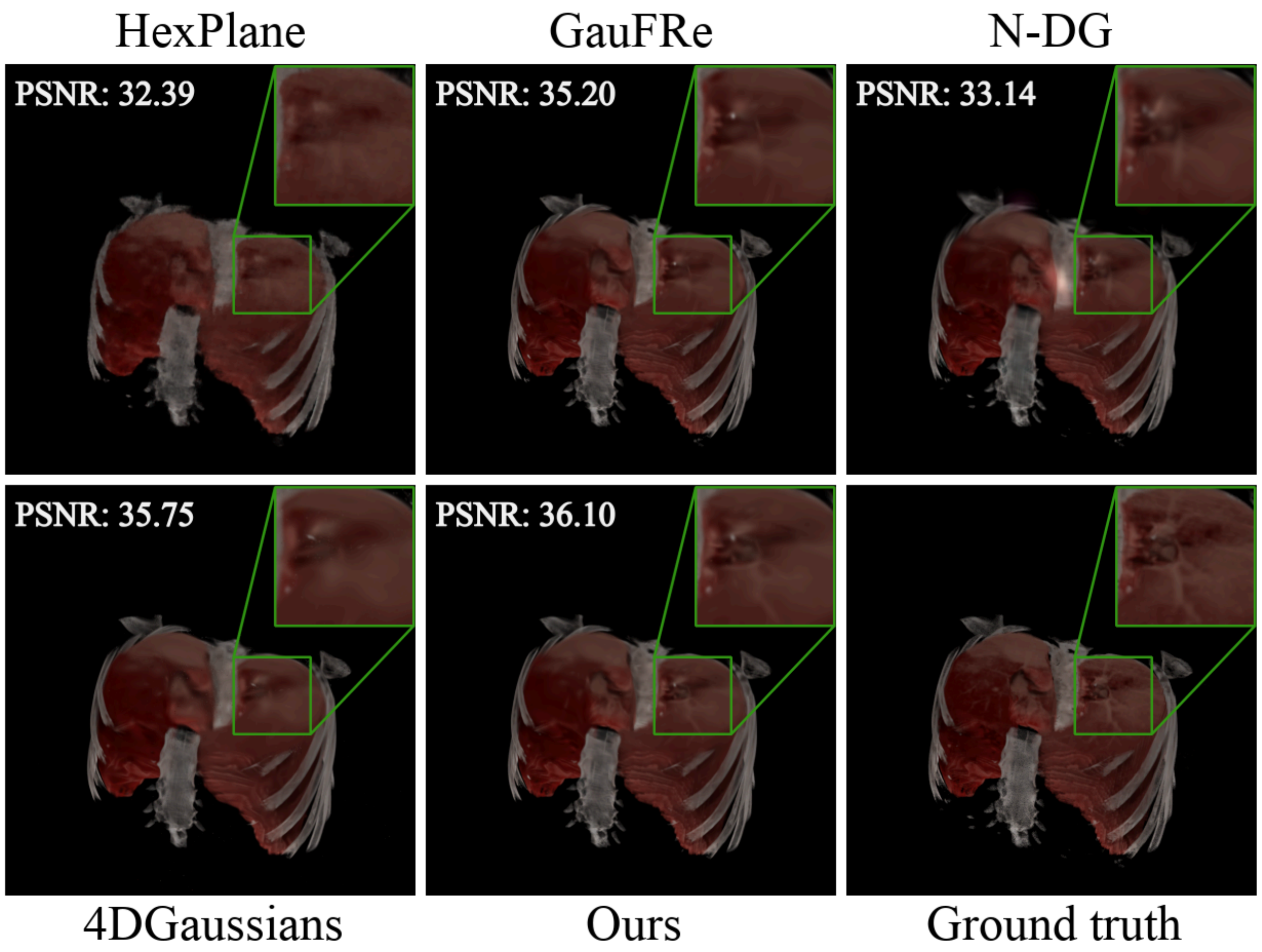
Methods for Comparison

- HexPlane: A Fast Representation for Dynamic Scenes
- GauFRe: Gaussian Deformation Fields for Real-time Dynamic Novel View Synthesis
- N-DG: N-Dimensional Gaussians for Fitting of High Dimensional Functions
- 4DGaussians: 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering

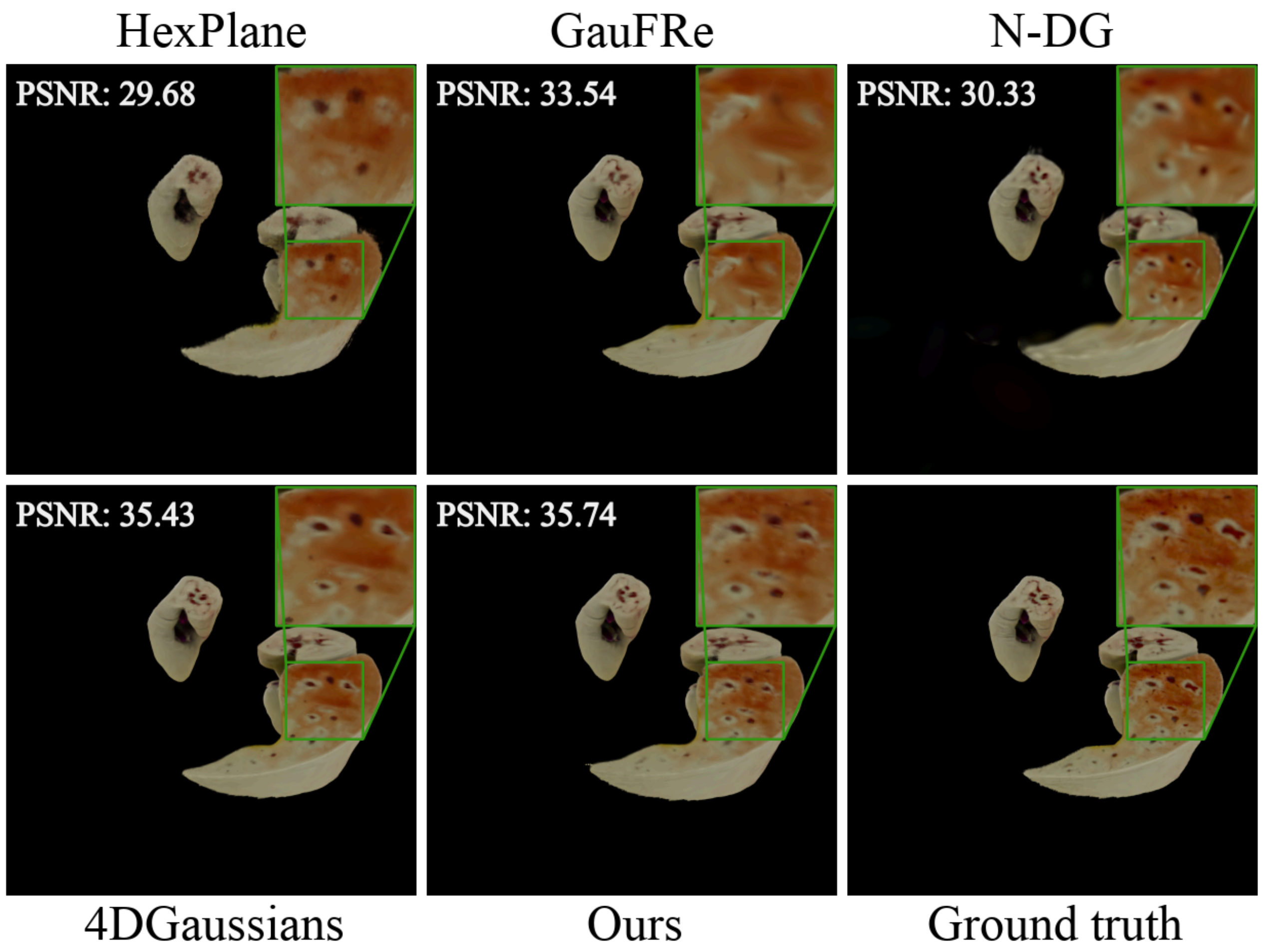
Comparison — Head



Comparison — Thorax



Comparison — Liver & Kidney



Comparison — Quantitative

Method	**PSNR↑**	**SSIM↑**	**LPIPS↓**	**Time↓**	**FPS↑**	**Storage↓**
HexPlane	31.836 ± 1.275	0.942 ± 0.015	0.109 ± 0.019	$14.1'$	2.7	68.0MB
GauFRe	34.717 ± 1.481	0.968 ± 0.009	0.062 ± 0.013	$16.6'$	150	15.0MB
N-DG	31.925 ± 1.541	0.937 ± 0.032	0.096 ± 0.015	$33.6'$	129	39.7MB
4DGaussians	35.653 ± 1.667	0.967 ± 0.009	0.079 ± 0.014	$9.7'$	87	17.8MB
Ours	36.635 ± 1.926	0.974 ± 0.010	0.061 ± 0.014	$13.7'$	156	16.1MB

Ablation Study

Method	PSNR↑	SSIM↑	LPIPS↓	Time↓	FPS↑	Storage↓
3DGS + LT	34.577 ± 1.441	0.969 ± 0.012	0.070 ± 0.015	$11.1'$	201	14.7MB
3DGS + AAM	35.946 ± 2.043	0.971 ± 0.013	0.064 ± 0.014	$11.5'$	148	15.6MB
3DGS + AAM + HT	36.255 ± 1.756	0.971 ± 0.008	0.063 ± 0.014	$14.1'$	150	15.7MB
3DGS + AAM + LT	36.635 ± 1.926	0.974 ± 0.010	0.061 ± 0.014	$13.7'$	156	16.1MB

Summary & Outlook

- Interactive cinematic rendering of medical data
- Extention of 3DGS
  - Learned Truncation
  - Adaptive Adjustment Model
- Medical cinematic dataset
- Incremental improvement

What's next?

- Clip plane direction?
- Even more natural clipping possible?